



DataPulse360

**Web-Based 360-Degree Feedback and Assessment
System for Management Education 2026
Requirements Document
Diana Livengood, Ivan Liang, Andreas Sofronas**

Table of Contents:

DataPulse360.....	1
Table of Contents:.....	2
Use Cases:.....	3
Student Self-Evaluation.....	3
Participant Feedback.....	3
Rater Submits Feedback.....	3
Introduction:.....	4
Objectives:.....	4
Functional Requirements:.....	5
- Administrator Management:.....	5
- Assessment Project Creation:.....	5
- Cohort Management:.....	5
- Monitoring Progress:.....	6
- As Participant:.....	6
- As Rater:.....	6
Nonfunctional Requirements.....	7
- Account restraints.....	7
- Performance.....	7
- Accessibility.....	7
Risk Assessment.....	7
- Security risks.....	7
- Data storage requirements.....	8
- Breach of anonymity.....	8
TimeLine:.....	8
Sprint 1 (9/10 - 9/17): Meeting with professor who requested the project.....	8
Sprint 2 (9/17 - 10/1): Setting up shell of the website.....	8
Sprint 3 (10/1 - 10/15): Functioning premade surveys for participants and raters.....	8
Sprint 4 (10/15 - 10/29): Backend survey creation for professor Jin.....	9
Sprint 5 (10/29 - 11/5): Report generation.....	9
Sprint 6 (11/5 - 11/19): Finalizing Ui design.....	9
Spring 7 (11/19 - 12/3): Testing and finalization.....	9
Anonymity and Confidentiality Rules:.....	10
Survey Configuration Requirements:.....	11
Data, Security, and Privacy:.....	12
Glossary.....	12

Use Cases:

Student Self-Evaluation

- **Goal:** Complete a self-evaluation before receiving feedback
- **Description:** The participant begins by logging into the system and opening the active assessment. The system then displays the assessment instructions for the participant to review. After reading the instructions, the participant answers all of the survey questions and submits the completed self-assessment. Once the assessment is submitted, the system validates the responses and stores them in the database. After successfully saving the responses, the system displays a confirmation message to the participant. Finally, the participant is returned to the dashboard.

Participant Feedback

- **Goal:** Participants will select evaluators who provide feedback
- **Description:** The participant begins by logging into the system and navigating to the Rater Management page. The participant then enters the email addresses of the evaluators, and the system validates the entered addresses. After the participant confirms the evaluator list, the system generates invitation records and sends invitation emails to the selected evaluators. Once the invitations have been sent, the participant receives a confirmation message. Finally, the dashboard is updated to display the list of invited raters.

Rater Submits Feedback

- **Goal:** The rater is able to submit anonymous feedback for a participant
- **Description:** The process begins when the rater receives an invitation email and clicks the unique survey link provided. The system then displays a consent page, where the rater reviews and accepts the consent agreement. After consent is given, the system displays the evaluation survey for the rater to complete. The rater answers all required questions and submits the

evaluation. Once the evaluation is submitted, the system stores the responses and deactivates the survey link to prevent additional submissions. Finally, the system displays a confirmation page indicating that the evaluation has been successfully completed.

Introduction:

DataPulse360 aims to design and develop a fully functional, secure, and configurable web-based 360-degree feedback system that supports management and leadership education. We seek to deliver a platform that allows instructors to run complete assessment cycles—from student registration to anonymous rater feedback to individualized reports—while ensuring flexibility for different survey instruments each semester. Our goal is to produce a reliable, scalable tool that can be used by Professor Sirkwoo Jin and potentially adopted by other departments. Users will be able to rate themselves on different leadership styles and skills, and receive peer based feedback on those same styles so they can see what their strengths and weaknesses are.

Objectives:

Our primary objective is to create a secure website that allows users to easily interpret and take self evaluation surveys based on their leadership and business skills. The surveys will take the form of a 1 to 5 rating system, with 1 being a crippling weakness and 5 being an amazing strength. These surveys will be stored securely so that outsiders cannot view them. After completing a survey students will invite at least three peers to take the survey, evaluating the original student's abilities as an outside party. This survey data will be collected into a centralized database, allowing us to compare the user's results to themselves, what their peers think of them, and the class as a whole. Feedback will be distributed through downloadable PDF documents, with the data being represented in terms of bar charts. The surveys themselves will be created by our admin users, professors, to

tailor the surveys to their classes specific needs.

Functional Requirements:

- **Administrator Management:**
 - Authentication:
 - Administrators shall be able to securely log in using authenticated credentials.
 - Administrators shall have access to an administrative dashboard.
 - Unauthorized users shall not be able to access administrator functions.
- **Assessment Project Creation:**
 - Administrators shall be able to create a new assessment project.
 - Administrators shall define: assessment title and description, start date, end date, survey template, and required minimum number of raters.
 - Administrators shall be able to modify assessment settings prior to release.
 - Administrators shall be able to create survey categories and questions.
 - Administrators shall be able to edit and delete survey questions.
 - Questions shall support multiple-choice and open-text responses.
 - Surveys shall be stored for future use.
- **Cohort Management:**
 - Administrators shall be able to create and manage multiple cohorts simultaneously.
 - Administrators shall be able to assign participants to specific cohorts.

-
- **Monitoring Progress:**
 - Administrators shall be able to view total participants, completed assessments, pending assessments, and the number of submitted rater evaluations.
 - Administrators shall be able to view incomplete assignments.

 - **As Participant:**
 - Students shall be able to log in using their school email accounts.
 - Students shall be able to enter the email addresses of more than three raters for their assessment.
 - The system shall send each rater a separate email containing a unique assessment link.
 - Students shall be required to complete their own assessment.
 - Students shall be able to log in to the student portal and view their assessment progress.
 - Students shall be able to select “Send Reminder” next to an individual pending rater.
 - Students shall be able to select “Send Reminder to All Pending Raters” to notify all pending raters simultaneously.

 - **As Rater:**
 - Raters shall be able to access an assessment through their unique assessment link.
 - Raters shall be presented with a consent screen before any survey items are displayed.
 - Raters shall complete the same survey as the student, with questions presented in the third person.
 - Raters shall be shown a brief confirmation screen after submitting the assessment.
 - A unique assessment link shall become inactive after the rater submits the assessment.
-

Nonfunctional Requirements

- Account restraints
 - Users will be required to sign up with their merrimack school accounts to ensure that our software is kept within house.
 - Administrator accounts will need to be separated from student accounts to prevent students from accessing confidential data and survey knowledge.
- Performance
 - The website should be responsive, will fast load times and fast data storage to prevent users from being confused.
 - Surveys must be made available to classes as close to their creation as possible.
 - The website must be able to support a minimum of 40 students at a time, for simultaneous survey taking during large classes.
- Accessibility
 - Our UI must be functional, easy to use, readable, and compatible with accessibility programs such as text to speech.

Risk Assessment

- Security risks
 - Since we were denied azure access for the Merrimack College portal, we will be unable to use Microsoft's secure login system. As we are not a team of microsoft cryptography experts any encryption system we implement for account protection will be inherently weaker.

-
- Data storage requirements
 - With multiple surveys per year, per class, the data storage requirements of our website is going to grow extremely quickly. Trimming this data will be vital to ensure ease of website use.
 - Breach of anonymity
 - Users should not be able to know what the people they invited rated them as. As such we will need to implement a minimum response requirement to ensure no data anonymity will be broken.

TimeLine:

Sprint 1 (9/10 - 9/17): Meeting with professor who requested the project

- *Exact* meaning of Survey Agnostic
- What does the professor want out of the comparison system
 - Number of questions per survey
 - Specifics of the rating system
- Formatting of the questions from the participant/raters PoV (ui design)
- Functionality Documents/Requirements
 - What is in the survey
 - What form will the survey take
 - How does the professor wish for us to edit the survey

Sprint 2 (9/17 - 10/1): Setting up shell of the website

- domain name,
- login system,
- mailing system for user authentication

Sprint 3 (10/1 - 10/15): Functioning *premade* surveys for participants and raters.

- Premade questions, categories, answers
- Rating system passed data that is stored in a spreadsheet somewhere

Sprint 4 (10/15 - 10/29): Backend survey creation for professor Jin

- Storing past surveys in a database
- Editable surveys
- Deletable surveys
- Create categories, questions, and answers

Sprint 5 (10/29 - 11/5): Report generation

- Spreadsheet report generation with privacy functions
- Fewer than X (3 default) responses = *null*
- Auto creating bar graphs for students to compare themselves with others

Sprint 6 (11/5 - 11/19): Finalizing Ui design

- Logos
- User friendly buttons
- Text rewrites for professionalism
- Ensuring mobile and desktop usability (lowkey low prio)

Spring 7 (11/19 - 12/3): Testing and finalization

- Fulfilling any missed/unfulfilled “bonus” requirements
- Testing with students
- Data retention time
- Finalizing everything with Professor Jin

Scoring and Feedback Analytics:

Survey items are grouped into theoretically defined constructs, and results are reported at the construct level rather than item by item. A leadership instrument might include constructs such as Emotional, Competency, Empowerment, Trust, Psychological Safety, and Transformational Leadership.

For each construct, the report should show at minimum:

- The participant's self-assessment score.
- The mean score across all other raters.
- The standard deviation of the other-rater scores, which shows how much raters disagreed.
- The number of raters included in the calculation.
- The difference between the self-rating and the mean other-rating.

Anonymity and Confidentiality Rules:

- Participants (i.e., students) may know whom they invited, but must never be able to determine which score or comment came from which rater.
- Individual rater responses never appear in a participant report. Only aggregated statistics are shown.
- Minimum-rater threshold: if fewer than the configured minimum number of raters (default three) have completed the assessment, the "Others" results are suppressed entirely rather than shown with a small n.

-
- If results are ever broken out by relationship type (supervisor, peer, subordinate), the same minimum-response rule must apply within each category — otherwise a single supervisor would be trivially identifiable.
 - The progress dashboard shows completion status only. A participant may see that three of five raters have finished, and which ones are still pending so reminders can be sent, but never any of their answers.
 - Administrative access is separate from response identity. I need to see participant and rater email addresses to manage the process, but rater identities should not be linked to individual responses in anything a participant can view.

Survey Configuration Requirements:

- The application must not be hard-coded to a single survey. I teach different courses with different validated instruments, and I need to be able to set up a new one myself without a developer.

A survey configuration should let me define, through the interface:

Construct → Survey Items → Response Scale → Scoring Rule → Report Output

For example: the construct Transformational Leadership contains 16 items with 4 dimensions, uses a 1–5 response scale, and its construct score is the average of the four dimension responses. Each item needs a self-referential wording (“I clearly explain my expectations”) and an other-referential wording (“This person clearly explains their expectations”), because the same construct is measured from both directions.

Data, Security, and Privacy:

- Authenticated accounts for administrators and participants, with passwords stored securely.
- Unique, single-purpose, expiring links for raters that grant access to exactly one assessment.
- Access controls prevent any participant from reaching another participant's data or reports.
- Separation of identifying information from response records wherever practical.
- The ability for me to delete an assessment project and all of its associated data once a course is over.

Glossary

- **360-Degree Feedback:** A multi-source feedback evaluation model where participants complete a self-assessment and receive peer-based feedback from multiple raters to identify personal strengths and areas for growth.
- **Administrator:** A system role—typically professors or instructors—authorized to create assessment projects, build survey instruments, assign cohorts, and monitor evaluation progress.
- **Cohort:** A designated group of students or participants assigned together to a specific assessment project for a given course or semester.
- **Minimum-Rater Threshold:** A privacy and confidentiality rule requiring a minimum number of completed evaluations (default is 3) before aggregate feedback is displayed, ensuring individual rater responses cannot be identified.
- **Participant (Student):** The primary user evaluated within the system who completes a self-assessment, invites raters, and views aggregated feedback reports.

-
- **Rater (Peer):** An individual invited by a participant to evaluate the participant's leadership and business skills through third-person survey questions.
 - **Standard Deviation:** A statistical calculation included in participant reports to show the variance and level of agreement among raters' scores for a given construct.
 - **Survey Agnostic:** A flexible architectural design that enables administrators to create, modify, and configure dynamic survey templates, scoring rules, and constructs without requiring source code changes.